

AJAY VIGNESHWARAN

AI / ML Engineer | Data Scientist | Data Analyst

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with 3+ years of experience designing, developing, and deploying machine learning and deep learning solutions across banking, automotive, and enterprise sectors. Proven ability to build end-to-end ML pipelines using TensorFlow and PyTorch, develop NLP models for text classification and entity recognition, and deploy models on cloud platforms (AWS). Adept at translating complex business challenges into scalable AI solutions with measurable impact. Experienced in MLOps practices, CI/CD pipeline development, and cross-functional stakeholder communication.

CORE COMPETENCIES

Deep Learning Frameworks: TensorFlow, PyTorch
NLP: Text Classification, NER, Text Summarization, Topic Modeling, SpaCy

Computer Vision: OpenCV, Object Detection, Image Classification

MLOps & Deployment: CI/CD for ML, REST APIs, AWS SageMaker, AWS EC2

Data Engineering: Python, SQL, PostgreSQL, MySQL, Feature Engineering

Visualization & BI: Tableau, Matplotlib, Seaborn

Libraries: NumPy, Pandas, Scikit-learn, Django

Cloud & DevOps: AWS, Git, GitHub, Docker

Statistical Methods: Predictive Modeling, EDA, Statistical Analysis

WORK EXPERIENCE

Senior Application Engineer (AI/ML Product Specialist)

CADD Centre Training Services
Chennai, Tamil Nadu

2023 – Present

- Designed and delivered end-to-end curriculum in Data Analytics, Python, and MLOps for 100+ corporate engineers at Tata Motors, Pune — demonstrating ability to bridge technical AI/ML concepts with real-world business requirements.
- Developed a comprehensive MLOps training module covering model lifecycle management, deployment pipelines, and AWS integration — directly aligned with CI/CD and cloud-based ML deployment.
- Conducted AI in Excel workshop at Madras Chamber of Commerce & Industry (MCCI) — showcasing AI solution evangelism and ability to communicate AI impact to non-technical stakeholders.
- Trained corporate professionals on AWS services including SageMaker, EC2, and S3 for ML workflows, reinforcing cloud AI/ML deployment skills.
- Ensured training quality through regular Facilitator Certification Programs, demonstrating leadership and continuous upskilling culture.

Associate Data Scientist

Skoruz Technologies
Bangalore, Karnataka

2022 – 2023

- Built a Bank Statement Analysis system processing 1000+ statements using statistical and ML models to assess creditworthiness of individuals and SMEs — end-to-end project ownership from data ingestion to model deployment.
- Performed large-scale data cleaning and feature engineering on 1,200+ customer records across banking and financial datasets using Python (Pandas, NumPy) — enabling higher-accuracy predictive models.
- Developed a Computer Vision model using deep learning (CNN) to detect helmet violations in traffic footage, achieving 80% accuracy — demonstrating real-world image recognition capability aligned with JD's vision/video deep learning requirements.
- Designed and developed ML models for 5+ research projects using end-to-end qualitative and quantitative data collection, supporting stakeholder presentations to 5+ project leads and 10+ non-technical audiences.

- Built and maintained database structures and schemas for 20+ business requirements using MySQL/PostgreSQL, supporting model pipelines with structured data access via SQL.
- Managed large datasets (10K+ observations) applying regex-based preprocessing and feature selection to build robust statistical models.

KEY PROJECTS & DELIVERABLES

Multiple Object Detection on Live Camera

- Developing a real-time object detection system using YOLO/OpenCV on live video feed — demonstrating computer vision and video analysis skills

Automated Car Parking System with Empty Slot Detection

- Implemented computer vision-based parking slot detection using image classification models — demonstrating applied deep learning on real-world spatial data.

Enhanced Document Verification using Hybrid Algorithm

- Published peer-reviewed paper at IJSART (Paper ID: IJSARTV6I637973, June 2020) on a hybrid algorithm for document verification — showcasing research, white paper writing, and knowledge dissemination aligned with JD's requirement to write white papers and create demonstrable AI assets.

CERTIFICATIONS

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| • Google Data Analytics — Google | • Data Analysis using Python |
| • Python for Data Science | • Python for Machine Learning |

EDUCATION

Bachelor of Engineering, Computer Science | KCG College of Technology, Chennai | 2016 – 2020